The operational use of drones in these early exercises provided critical empirical data on flight control reliability, radio communication stability, and human-machine coordination. It also allowed defense organizations to refine their approaches to range control, safety protocols, and weapon calibration.

2.9 Summary

The chapter represents the formative stage in the evolution of unmanned aerial vehicles (UAVs), spanning from the early 1910s to the late 1930s. This era marked the transition from conceptual experimentation to practical application, driven by wartime innovation and early advances in radio control, aerodynamics, and propulsion. Early UAVs were manually operated, fixed-wing aircraft with simple mechanical linkages, lightweight wooden or canvas airframes, and basic radio systems, lacking sensors or autonomy. The foundations of unmanned flight emerged through pioneering concepts such as tethered balloons, kites, and wireless control experiments by innovators like A. M. Low, Charles Kettering, and Reginald Denny. Key developments included the *Aerial Target (1916)* and *Ruston Proctor Aerial Target (1917)* in the United Kingdom, the *Kettering Bug (1918)* in the United States, and later operational drones such as the *Queen Bee (1935)* and *Radioplane OQ-2 (1939)*. These systems were primarily used for gunnery training, reconnaissance, and experimental bombing, showcasing the potential of unmanned flight in defense operations. Despite limitations in range, stability, and control precision, first-generation drones laid the technological and conceptual groundwork for modern UAV design, establishing key principles in radio communication, automation, and aerial control that would guide future developments in military and civilian drone technology.